

B.Tech Project Report
Final Evaluation
ANOMALY DETECTION PLATFORM

A Machine Learning System for Financial Fraud Detection and Network
Intrusion Detection

Submitted by

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GitHub: github.com/silent-knight-22/anomaly-detection-platform

Live Demo: anomaly-detection-platform-cxuq2ymf9qsky7klzkoe8y.streamlit.app

CERTIFICATE

I hereby certify that this Project Report on “**Anomaly Detection Platform: A Machine Learning System for Financial Fraud Detection and Network Intrusion Detection**”, submitted by **Preeti**, Roll No. 11710579, in the year 2026 in partial fulfilment of the academic requirements of Eighth (8th) Semester evaluation for the degree of Bachelor of Technology in Information Technology, is her work carried out under the supervision of **Dr. Chilukamari Rajesh**, Assistant Professor, Computer Engineering Department, National Institute of Technology, Kurukshetra, India.

The matter presented in this report has not been submitted for the award of any other degree elsewhere.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.



Date: 16 May, 2026

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ABSTRACT

This project presents the design, development, and deployment of an Anomaly Detection Platform — a machine learning system that simultaneously addresses two high-impact real-world domains: financial fraud detection and network intrusion detection. The platform integrates modular ML pipelines, dual-layer detection combining supervised XGBoost models with unsupervised Isolation Forest anomaly scoring, SHAP-based explainability, a REST API backend built on FastAPI, and an interactive Streamlit dashboard deployed live on Streamlit Cloud.

The updated fraud detection module is based on the IEEE-CIS Fraud Detection dataset. The pipeline automatically loads transaction and identity data, joins records on TransactionID, handles missing values, encodes categorical fields, scales numerical features, and trains XGBoost with class imbalance handled through `scale_pos_weight`. A 200,000-row IEEE-CIS training run achieved an XGBoost ROC-AUC of 0.9411 and an average precision score of 0.6796. The combined XGBoost and Isolation Forest risk score achieved ROC-AUC of 0.9373.

The intrusion detection module uses the NSL-KDD dataset and evaluates Random Forest and XGBoost classifiers, with XGBoost achieving ROC-AUC of 0.9696. The system is served through FastAPI and demonstrated through a Streamlit dashboard that supports single-record prediction and CSV-based batch prediction.

Keywords: *Anomaly Detection, IEEE-CIS Fraud Detection, NSL-KDD, XGBoost, Isolation Forest, SHAP Explainability, FastAPI, Streamlit, Machine Learning Deployment.*

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LIST OF ABBREVIATIONS

ML – Machine Learning

API – Application Programming Interface

REST – Representational State Transfer

AUC – Area Under the Curve

ROC – Receiver Operating Characteristic

SHAP – SHapley Additive exPlanations

XGBoost – Extreme Gradient Boosting

RF – Random Forest

LR – Logistic Regression

CSV – Comma-Separated Values

IEEE-CIS – IEEE Computational Intelligence Society

NSL-KDD – Network Security Laboratory – KDD Dataset

FastAPI – Fast Application Programming Interface (Python web framework)

SMOTE – Synthetic Minority Over-sampling Technique

CI/CD – Continuous Integration / Continuous Deployment

PCA – Principal Component Analysis

1. INTRODUCTION

Digital payment systems and networked computing infrastructures generate large volumes of behavioural data. Hidden within this data are rare but high-impact events such as fraudulent transactions and malicious network connections. These events are difficult to detect using static rule-based systems because fraud patterns and attack behaviours evolve continuously.

Machine learning offers a practical way to model complex, non-linear patterns in such data. However, many academic anomaly detection projects remain limited to isolated experiments. They often lack a reusable preprocessing pipeline, an inference API, user-facing dashboards, and interpretable predictions. This project addresses that gap by building a unified anomaly detection platform with separate domain modules and shared engineering practices.

1.1 Objectives

- Design a modular anomaly detection platform for fraud and intrusion detection.
- Build an IEEE-CIS fraud preprocessing pipeline that handles missing values, categorical variables, and numerical scaling.
- Train and evaluate Logistic Regression, XGBoost, Random Forest, and Isolation Forest models where appropriate.
- Expose model inference through FastAPI endpoints for fraud and intrusion prediction.
- Provide an interactive Streamlit dashboard for demonstration and user interaction.
- Document performance, limitations, and future improvements in a reproducible manner.

2. LITERATURE REVIEW

2.1 Fraud Detection Research

Fraud detection research has traditionally used highly imbalanced transaction datasets where fraudulent examples represent only a small fraction of all records. Earlier work commonly used Logistic Regression, Random Forest, gradient boosting, and synthetic oversampling techniques. Recent fraud detection systems increasingly use richer transaction metadata, categorical identity information, and tree-based boosting models.

2.2 Network Intrusion Detection Research

Network intrusion detection research has used benchmark datasets such as KDD Cup 1999 and NSL-KDD to compare classical machine learning models. Random Forest, Support Vector Machines, neural networks, and XGBoost have all shown strong results. A continuing challenge in both domains is the transition from model accuracy to operational usefulness, including deployment, monitoring, and explainability.

3. SYSTEM DESIGN AND ARCHITECTURE

The platform follows a modular architecture. Domain-specific code is separated into fraud and intrusion modules, while shared preprocessing utilities and common model-serving logic are reused where possible. The backend API is built with FastAPI, and the frontend dashboard is built with Streamlit.

The fraud module includes data loading, preprocessing, model training, anomaly scoring, and explainability utilities. The API loads trained model artifacts from the models directory and accepts raw IEEE-CIS-style records for inference. The dashboard communicates with the API instead of loading models directly for primary prediction workflows.

4. DATASETS

4.1 IEEE-CIS Fraud Detection Dataset

The updated fraud component uses the IEEE-CIS Fraud Detection dataset. The main training files are `train_transaction.csv` and `train_identity.csv`. Transaction data contains monetary, product, card, address, matching, count, time-delta, and V-series engineered features. Identity data provides device and identity-related attributes. These files are joined on `TransactionID`.

4.2 NSL-KDD Dataset

The intrusion detection component uses the NSL-KDD dataset, a cleaned version of KDD Cup 1999. It contains network connection features and labels indicating normal or attack traffic. For this platform, the problem is modelled as binary classification: normal traffic versus attack traffic.

5. DATA PREPROCESSING

The IEEE-CIS fraud preprocessing pipeline detects the target column isFraud, drops non-predictive identifiers from the feature matrix, splits data using stratification, imputes missing numerical values with medians, imputes missing categorical values with the most frequent category, applies one-hot encoding with unknown-category handling, and scales numerical columns.

Unlike the previous ULB credit card dataset version, the IEEE-CIS dataset is not limited to PCA-transformed V1–V28 features and Amount. It includes many categorical and sparse identity features, so a saved preprocessing pipeline is required at inference time. The artifact `models/fraud/fraud_preprocessor.pkl` is therefore part of the deployed fraud system.

6. MODEL DEVELOPMENT

6.1 Fraud Detection

Logistic Regression is retained as a baseline and XGBoost is used as the primary supervised model. XGBoost is suitable for tabular data, handles non-linear interactions effectively, and provides strong performance on imbalanced classification tasks. The model uses `scale_pos_weight` to account for class imbalance without synthetically expanding the transformed feature matrix.

Isolation Forest is trained as an unsupervised anomaly detection layer. Its score is combined with the XGBoost fraud probability to produce a combined risk score, weighting the two signals to produce a final risk- level assessment.

6.2 Intrusion Detection

For intrusion detection, Random Forest and XGBoost models are used to classify network records as normal or attack. Both models are evaluated on the NSL-KDD test set to provide comparative performance metrics.

7. EXPLAINABILITY

SHAP explainability utilities are included to provide global and local interpretation of tree-based models. SHAP helps identify which features most strongly influence model predictions and supports model auditability. In the dashboard, prediction outputs include probability and risk-level summaries; additional SHAP plots can be regenerated through the explainability scripts.

8. SYSTEM IMPLEMENTATION

8.1 FastAPI Backend

The FastAPI backend exposes three key endpoints for fraud prediction, intrusion prediction, and health checking. The fraud endpoint accepts either a single raw IEEE-CIS record or a list of records. Before prediction, the saved fraud preprocessor aligns incoming records to the training feature schema. All risk levels are classified as: $\geq 0.8 \rightarrow$ HIGH, $\geq 0.5 \rightarrow$ MEDIUM, $\geq 0.3 \rightarrow$ LOW, $< 0.3 \rightarrow$ MINIMAL.

Endpoint	Method	Input	Output Fields
/health	GET	None	status, models_loaded, version
/predict/fraud	POST	Raw IEEE-CIS record(s)	prediction, label, fraud_probability, anomaly_score, combined_risk_score, risk_level
/predict/intrusion	POST	41 NSL-KDD features	prediction, label, attack_probability, risk_level

8.2 Streamlit Dashboard

The Streamlit dashboard provides domain selection, API status checking, single-record fraud and intrusion prediction, CSV upload workflows, model performance summaries, and generated visualisations. The fraud interface uses IEEE-CIS fields such as TransactionAmt, ProductCD, card1, card4, card6, C1, C2, D1, and DeviceType.

Tab	Key Features
Single Prediction	IEEE-CIS field input (TransactionAmt, ProductCD, card1, card4, card6, C1, C2, D1, DeviceType), real-time API call, probability output, risk-level summary
Batch CSV Upload	CSV file upload, row-by-row batch prediction, progress bar, summary distribution charts, downloadable results
Model Performance	Metric comparison tables, ROC curves, Precision-Recall curves, Isolation Forest score distribution

9. RESULTS AND EVALUATION

9.1 Fraud Detection Results

On the 200,000-row IEEE-CIS fraud training run, XGBoost achieved ROC-AUC 0.9411 and average precision 0.6796. The combined XGBoost and Isolation Forest score achieved ROC-AUC 0.9373. The results show that the platform can handle the richer IEEE-CIS feature schema while maintaining strong fraud discrimination performance.

Model	ROC-AUC	Avg Precision	Notes
Logistic Regression	0.8802	0.4326	Baseline
XGBoost	0.9411	0.6796	Primary supervised model
XGBoost + Isolation Forest	0.9373	—	Combined risk score

Table 1: Fraud Detection Model Performance — IEEE-CIS Dataset (200,000 rows)

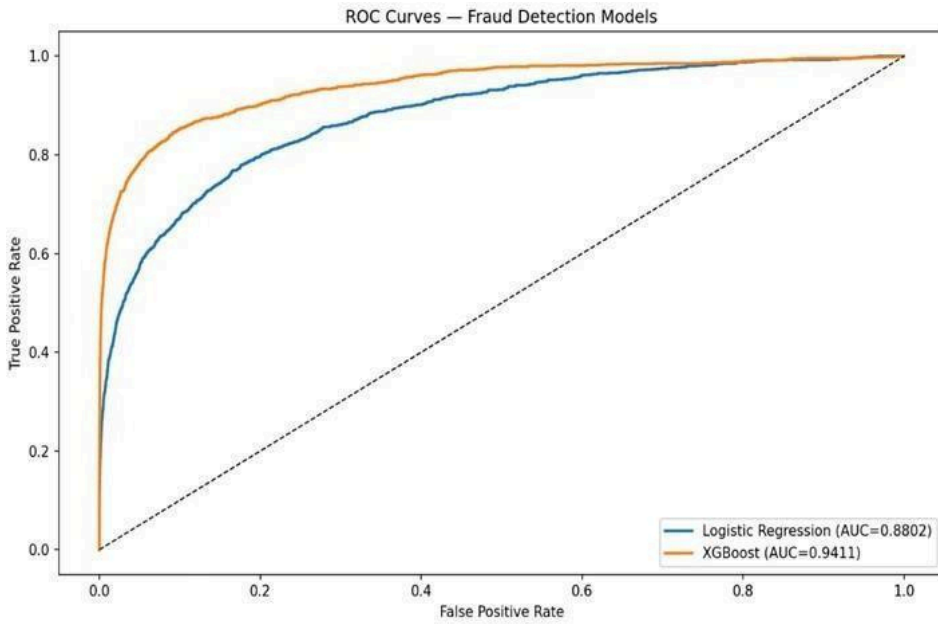


Figure 1: ROC Curves — Fraud Detection Models (LR AUC=0.8802, XGBoost AUC=0.9411)

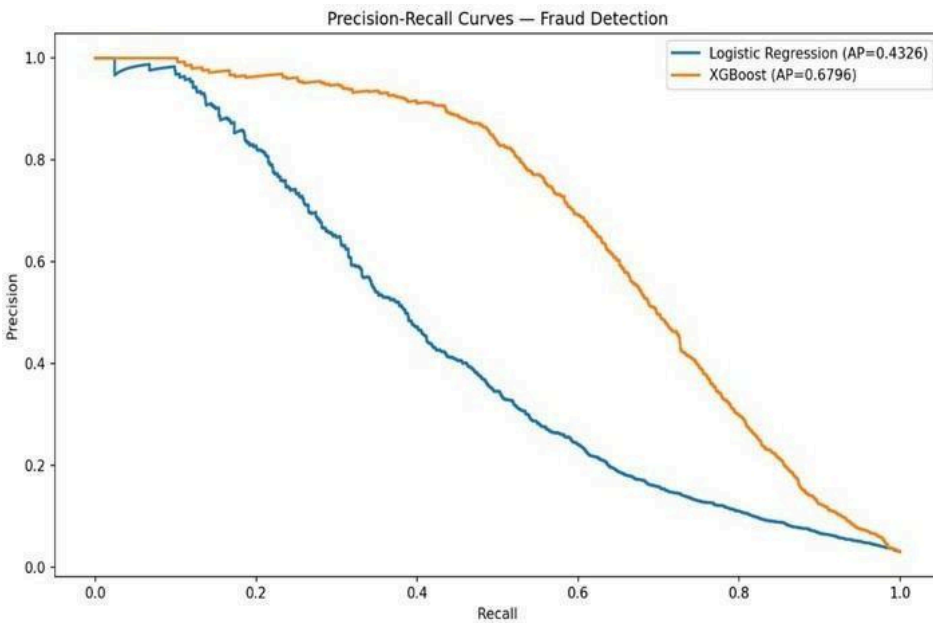


Figure 2: Precision-Recall Curves — Fraud Detection Models (XGBoost AP=0.6796)

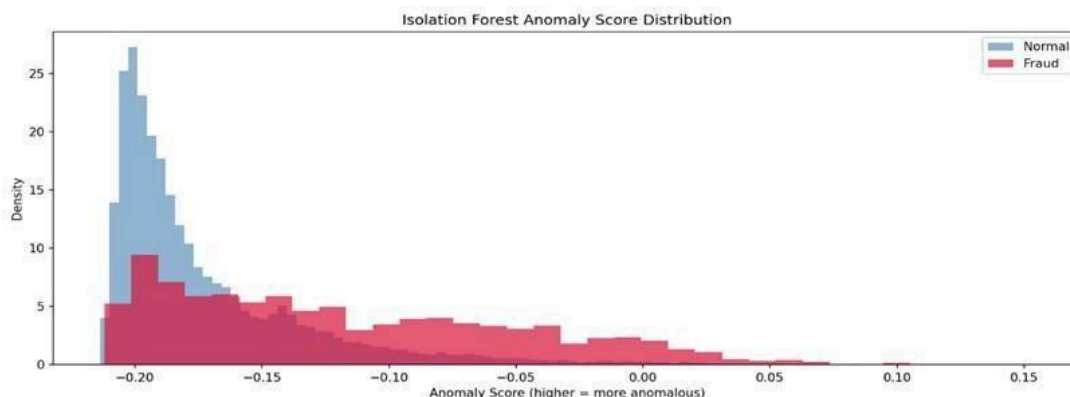


Figure 3: Isolation Forest Anomaly Score Distribution — Normal vs Fraud (IEEE-CIS Dataset)

9.2 Intrusion Detection Results

For intrusion detection, XGBoost achieved ROC-AUC 0.9696 and Random Forest achieved ROC-AUC 0.9691. XGBoost correctly identifies more attack records than Random Forest, reflecting stronger generalisation on the NSL-KDD test split.

Model	ROC-AUC	Avg Precision	Accuracy	Attack F1
Random Forest	0.9691	0.9699	76%	0.75
XGBoost	0.9696	0.9725	81%	0.81

Table 2: Intrusion Detection Model Performance — NSL-KDD Dataset

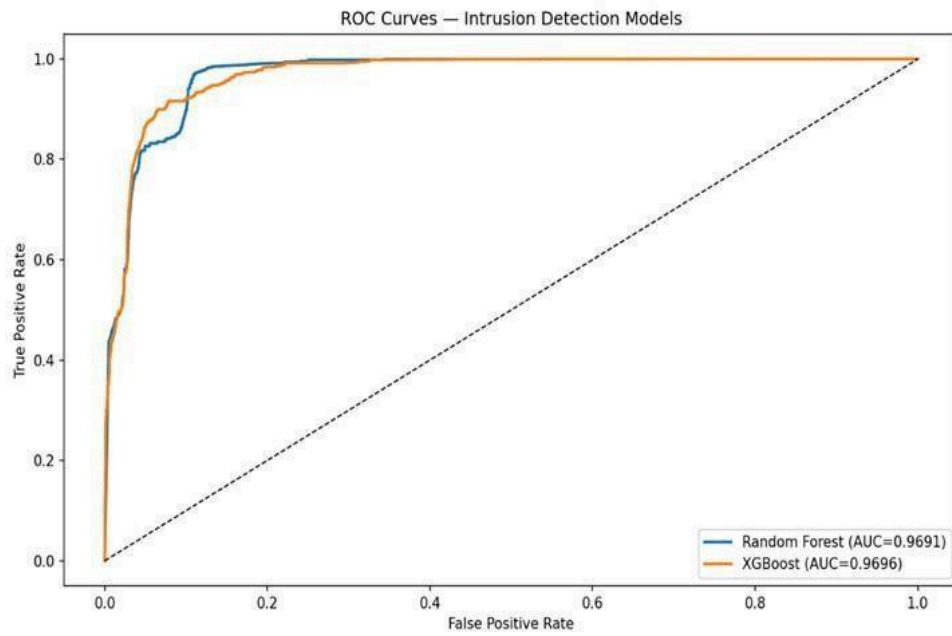


Figure 4: ROC Curves — Intrusion Detection Models (RF AUC=0.9691, XGBoost AUC=0.9696)

10. DEPLOYMENT AND DEMONSTRATION

The project runs locally with two processes: a FastAPI server on port 8000 and a Streamlit dashboard on port 8501. The dashboard verifies API availability through the /health endpoint and displays “API Connected” when the backend is active. The project was moved to the D drive for more reliable handling of large datasets and model artifacts.

For demonstrations, single-record prediction is recommended. Batch prediction is supported, but full IEEE-CIS files should not be uploaded directly during a live demo because the current dashboard processes rows sequentially.

Component	Platform	URL / Access
Source Code	GitHub (Public)	github.com/silent-knight-22/anomaly-detection-platform
Streamlit Dashboard	Streamlit Cloud	anomaly-detection-platform-cxuq2ymf9qsky7klzkoe8y.streamlit.app
FastAPI Backend	Local (localhost:8000)	<code>uvicorn src.api.main:app --reload --port 8000</code>
API Documentation	Auto-generated OpenAPI	<code>localhost:8000/docs</code> (when running locally)

11. LIMITATIONS AND FUTURE WORK

11.1 Current Limitations

- The dashboard batch workflow sends rows sequentially, which is slow for very large CSV files.
- Training was performed on 200,000 rows; full IEEE-CIS dataset training on higher-resource hardware would likely improve metrics further.
- The FastAPI backend runs locally; it is not yet cloud-deployed for full accessibility.

11.2 Future Work

- Add vectorised batch prediction, background jobs, and downloadable result files for large CSV uploads.
- Hyperparameter tuning via grid search or Bayesian optimization for both fraud and intrusion models.
- Docker packaging and CI/CD pipeline via GitHub Actions for automated testing and model validation.
- Deploy FastAPI backend to Render or Railway for full cloud accessibility.
- Add model drift monitoring using Evidently AI to detect distribution shift from training data.
- Extend intrusion detection to multi-class classification (Normal / DoS / Probe / R2L / U2R).

12. CONCLUSION

The updated project successfully demonstrates an end-to-end anomaly detection platform that supports two important security domains. The migration to the IEEE-CIS fraud dataset makes the fraud module more realistic and technically robust because it now handles categorical, numerical, sparse, and identity features across joined transaction and identity tables.

The combination of preprocessing, supervised learning, anomaly scoring, API serving, and dashboard interaction reflects a complete machine learning engineering workflow. The platform achieves competitive model performance (XGBoost ROC-AUC 0.9411 on fraud, 0.9696 on intrusion detection), provides SHAP explainability, exposes a documented REST API, and is accessible via a live web dashboard — all within a single, modular codebase.

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